Part II – Introduction

Part I established the conceptual foundations guiding this project: Decision Support Systems theory defined the conditions under which technology can bridge raw data and actionable decisions; the Technology Acceptance Model identified the behavioral factors that determine tool adoption in agricultural contexts; and precision agriculture provided the operational framework within which data-driven crop management generates measurable outcomes. These conclusions converge on a single applied question: can a localized, accessible decision-support system constitute a viable response to the decision-making deficit structuring Algerian smallholder agriculture? Part II is constructed to answer this question across five chapters — from technical feasibility to financial viability.

1 Origin of the Idea

1.1 Direct Observational Trigger

AGRIGENT originated from direct field observation were productivity fell despite heavy fertilization. Applications were found to be based on habit, not soil needs, resulting in poor outcomes and soil damage from uncontrolled chemical accumulation. This pattern was not isolated. Engagement with other farmers in the same area confirmed that irrigation and fertilization decisions are routinely made through visual observation and generalized practice rather than soil-specific data.

1.2 The Investor Failure Case

A second, more consequential case reinforced this diagnosis a new agricultural investor initiated an agricultural project without prior agronomic experience, relying on informal guidance. No soil analysis was conducted; no irrigation schedule was designed. fertilization was applied routinely ignoring crop and soil conditions. The project failed within its first cycle.

the two observations — both cases share a one cause: the gap between available soil data and the farmer's decision-making process is not bridged by any accessible, localized, real-time tool.

2 Problem Definition

Algeria's agricultural sector contributes 13.09% of GDP and employs 9.34% of the workforce (World Bank, 2023), yet its productive performance remains weak. Across 8.5 million hectares of agricultural land, wheat yields stand at only 1.5 - 1.7 tons per hectare — well below comparable countries under similar conditions (FAO, 2020) — while fertilizer application average just 20.7 kg/h (FAO/World Bank, 2023), among the lowest in the region. low input use and low yields together indicate not a resource scarcity problem, but a resource management problem. The underlying cause is information: The key agricultural decisions that directly influence productivity — irrigation, fertilization, and crop selection — are often made without real soil data. Instead, they rely on estimation, past habits, and generalized advice. This leads to three main consequences: Inefficient use of water and fertilizer, reduced ability to diagnose and correct crops, increased production costs with limited gains in productivity.

3 Objectives, Product, and Value Proposition

agrigent aims to provide farmers and agricultural investors with a real-time decision-support tool that transforms soil data into actionable agronomic guidance, reducing reliance on estimation in irrigation, fertilization, and crop management.

The system combines soil sensors (moisture, NPK, pH) with a mobile application that analyzes data with artificial intelligence and delivers clear recommendations adapted to local conditions.

AGRIGENT creates value by improving decision accuracy, reducing input waste, and lowering operational risk. It offers a simple, affordable, and locally adapted alternative to traditional advisory services, with real-time field accessibility.

1 Chapter 1: Technical Study

1.1 Functional Analysis and Solution Design

1.1.1 General presentation of the solution

Nature and identity of the system

AGRIGENT is a hardware-software agricultural decision-support system designed to assist Algerian farmers and agricultural investors in making data-informed irrigation, fertilization, and crop management decisions. The system integrates a field-deployable soil sensor unit with a mobile application that translates measured soil parameters into prioritized agronomic recommendations calibrated to local crop types and climate conditions.

a. Prototype: a functional implementation of core system architecture. It is rule-based engine designed to demonstrate technical feasibility and validate the recommendation logic under real field conditions.

b. full system: the commercial product to which is being built. It integrates complete sensor coverage, a fully populated agronomic database, and a machine learning extension and artificial intelligence layer for predictive decision support.

Problem addressed

The central operational problem AGRIGENT addresses is the absence of a real-time information channel between measurable soil conditions and field-level management decisions. Currently, critical tasks—including irrigation timing, fertilization rates, and crop suitability assessments—rely on estimation and generalized habits rather than site-specific data. AGRIGENT bridges this gap by providing a practical, low-cost, Arabic-language decision-support tool designed for users without formal agronomic training.

Primary objectives of the solution

- Improve the accuracy of agricultural decision-making through data-driven insights.
- Enhance crop productivity by aligning practices with actual soil requirements.
- Optimize the use of water and fertilizer resources under local farming conditions.
- Accumulate structured data for future development and decision support.

Target users and context of use

The system is designed for three user profiles operating in Algerian agricultural conditions

a. **farmers and experienced producers** managing active crop production, seeking to optimize input use and reduce yield uncertainty.
b. **New agricultural investors** without specialized agronomic training, requiring structured decision support to manage production risk.
c. **Agricultural cooperatives** seeking scalable soil monitoring capacity across multiple fields or zones.

As for the operational context, it is field based, where users interact with the system through a mobile application. Soil readings are initiated on demand by activating the sensors unit, which transmits data to the application via Bluetooth, short-range communication or Wi-Fi.

The system is designed to operate under conditions of weak internet connectivity. The core function — data acquisition — is performed locally on the device, ensuring continuous collection of the data in rural agricultural environments with limited network access.

Positioning relative to existing solutions

Existing local systems are 'standalone' devices that provide raw measurements without data processing, interpretation, or decision-support capabilities on simple screens, forcing users to perform their own analysis. AGRIGENT shifts toward 'Decision Support Systems' (DSS) transforming data into direct actionable recommendations. And unlike traditional hardware, it ensures 'digital data storing' for historical tracking and offers a localized interface specifically designed for the Algerian farmer's needs.

The following table provides a detailed technical comparison.

Table 1-1Positioning relative to existing solutions

Technical Criterion	Local Devices	International solutions	AGRIGENT
Data Processing	Raw data	Advanced analytics	Locally Calibrated DSS
Connectivity	None (Isolated)	Cloud-dependent	Cloud-dependent
Historical Data	No	Yes	Yes
Language Support	English Only	No Arabic	Arabic / Localized UI
System Architecture	Hardware-only	SaaS / IoT platforms	DSS / sensor integration
Decision Support	Not available	Yes	Yes

Source: Compiled by the author.

1.1.2 Requirements Analysis

a. Functional requirements

AGRIGENT's full system provides five core functional features, organized by system layer. The table below describes each feature as designed for the complete system.

Table 1-2 Functional requirements

Feature	Layer	Full System Description
soil moisture measurement and transmission	Hardware	Continuous on-demand field measurement with wireless data transmission to the mobile application
Soil pH measurement and transmission		
Soil NPK composition measurement and transmission		
recommendation engine (irrigation, fertilization, pH, crop suitability)	Software Hardware-only	Full agronomic coverage across all crop types, growth stages, and Algerian climate zones
Farmer outcome feedback module		data analyses and recommendations to farmer

Source: Compiled by the author.

User Profiles and Access Rights

The system defines tow access profiles, each with distinct interaction rights and data visibility boundaries.

Farmer / Investor (primary user)

The system shall allow primary users to create and manage a personal agronomic profile specifying wilaya, crop type, and active growth stage. Users shall initiate on-demand soil readings, receive prioritized recommendations linked to their profile, and access a chronological log of past readings and generated recommendations. users shall submit seasonal yield and outcome feedback.

System Administrator

The system shall provide administrators with tools to manage agronomic database content, update parameter ranges, monitor system-level performance indicators, and manage engineer partner accounts. Administrative access shall not extend to individual user field data.

User Journeys

Two primary journeys define the operational flow of the full system.

a. **Journey 1 — In-Season Field Management (primary use case)**
b. **Journey 2 — Pre-Planting Crop Suitability Assessment**

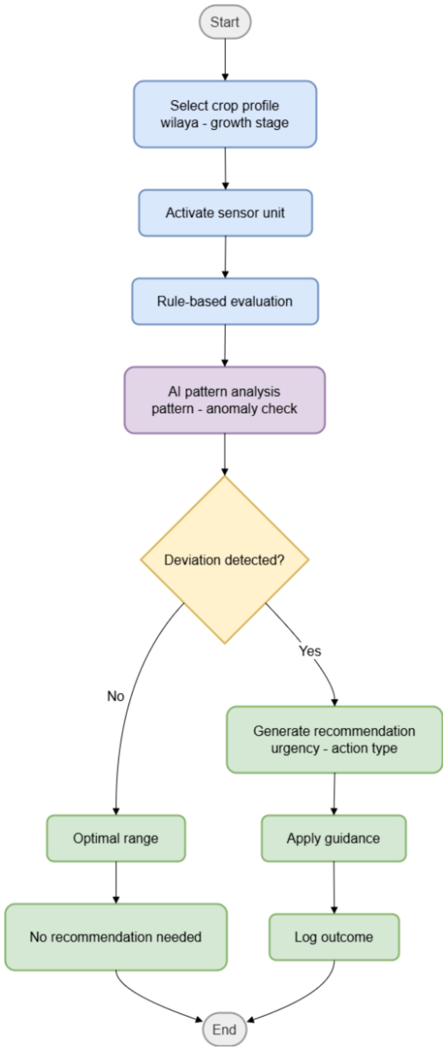


Figure 1-1Journey 1 — In-Season Field Management

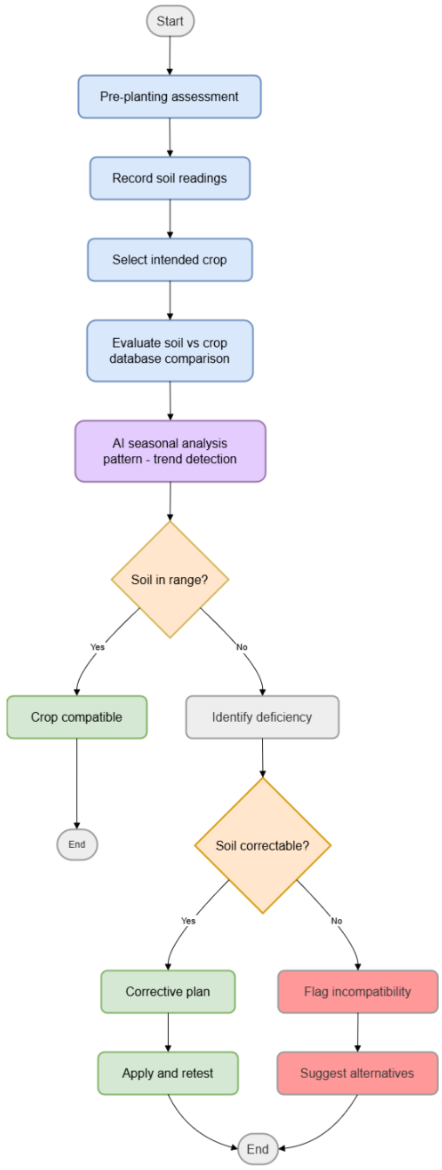


Figure 1-2Journey 2 — Pre-Planting Crop Suitability Assessment

Business Processes and Data Flows

The system shall process soil data through four sequential stages, from physical measurement to actionable output.

Figure 1-3Business Processes and Data Flows

Stage 1 — Acquisition

The sensor unit measures soil moisture, NPK composition, and ph. The measurements are sent to the microcontroller ESP32 to be converted into electrical signals and digitized. The digitized values are transmitted wirelessly to the user's smartphone.

Stage 2 — Reception

The mobile application receives the values from the device and links them with the user's agronomic profile — crop type, growth stage, and climate zone — which serves as the main key for the evaluation stage.

Stage 3 — Evaluation

The application checks the farming database using the information from the current profile. For each received value, the system determines whether the reading falls within, below, or above the validated optimal range for the corresponding crop-stage-zone combination. The evaluation logic is rule-based and deterministic in the current implementation; Phase 2 extends this layer with a machine learning module trained on accumulated field outcome data.

Stage 4 — Output

The system generates clear and structured recommendations for the user. For each factor, it indicates whether the value is within, below, or above the optimal range. When a deviation is detected, the system highlights it and provides a prioritized recommendation, along with a general corrective action based on the level of the deviation (mild, moderate, or significant).

b. Non-Functional Requirements

Performance

The system shall generate a complete recommendation output within five seconds of receiving a sensor reading under normal operating conditions.

Scalability

The system architecture shall support expansion along two dimensions:

Database scalability: new crop types, growth stages, climate zones, and agronomic parameters shall be added without restructuring the existing database schema.

Infrastructure scalability: Server-side components supporting data aggregation shall be designed to accommodate user base growth.

Security and Data Confidentiality

User agronomic profiles and field reading data constitute commercially sensitive information. The system shall implement:

- Encrypted local storage of user profiles, reading history, and recommendation logs.
- Secure user authentication and access control.
- Encrypted data transmission (HTTPS/TLS) for all server communications.
- Compliance with Algerian Law No. 18-07 (2018) on personal data protection.

Accessibility and User Experience

The system shall be designed for ease of use under real field conditions:

- Arabic shall be the primary interface language.
- Recommendation shall be Clear, color-coded outputs with simple language.
- The application shall be compatible with Android 8.0 and iOS 13 and above, covering most smartphones in active use.
- The interface shall be designed for legibility under outdoor light conditions.

Maintainability and Evolvability

The system shall be structured to support ongoing development without architectural disruption:

- Agronomic database updates shall be deployable easily.
- Machine Learning Module, enhanced feedback system shall integrate as extensions to the existing architecture, not as replacements of core components.

1.2 Technical Architecture and Implementation

1.2.1 Global Technical Architecture

Architectural Model

AGRIGENT adopts client-dominant distributed architecture, in which the mobile application functions as the primary processing environment. The application hosts both the recommendation engine and the local agronomic database, enabling full operational independence from network connectivity. The backend server layer supports secondary functions — outcome data synchronization and AI model updates — but is not required for core system operation. This architectural choice is not incidental: it is a deliberate response to the connectivity conditions.

The architecture is organized across five functional layers:

Layer 1 — Hardware (Sensor Unit)

The sensor unit is the physical interface between the agricultural environment and the digital system. It consists of three sensors — soil moisture, NPK composition, and soil pH — connected to an ESP32 microcontroller. The unit is field-deployable, battery-powered, and weather-resistant.

Layer 2 — Data (Local Agronomic Database)

The agronomic reference database is stored locally within the mobile application. It contains validated optimal parameter ranges for soil, organized by crop type, growth stage, and Algerian climate zone, as described in Section 1.1.2. The recommendation engine queries this database synchronously at the time of each soil reading evaluation.

Layer 3 — Backend / Service

The backend provides two services. First, it stores the anonymized and aggregated outcome dataset collected from the mobile application — soil readings, generated recommendations, and farmer-reported results — which constitutes the training corpus for AI models. Second, it hosts updated versions of the agronomic database, which are pushed to devices upon connection.

Layer 4 —Artificial intelligence

The AI layer is a backend-side service that becomes operational once the outcome dataset reaches sufficient volume and geographic diversity — estimated at 18 to 24 months of deployment across multiple wilayas. It performs three analytically distinct functions:

- **Yield risk classification:** Using an XGBoost classifier, a model trained on soil data and yield history this model predicts yield risk (low to high) from seasonal soil profiles, crop types, and climate data. It was specifically chosen for its effectiveness in processing structured agricultural data with limited training volumes.
- **Recommendation refinement:** This model refines rule-based outputs by comparing them to observed farmer outcomes. Using collaborative filtering, it adjusts generic guidance based on the performance of similar agronomic profiles.
- **Soil degradation detection:** This model identifies flags soil readings deviating from historical norms for a specific field and season. Model type: isolation forest or autoencoder, depending on training data volume at deployment.

Layer 5 — Application (Mobile Application)

The mobile application is the operational core of the system. It performs four functions: receiving and parsing sensor data transmitted from the hardware unit; querying the local agronomic database using the user's active crop profile as lookup parameters; executing the recommendation engine logic; and presenting structured decision outputs to the user. The application is cross-platform (Android and iOS), built for offline-first operation, and stores all user profiles, sensor readings, and recommendation history locally on the device.

Data Flow

The complete data flow through all five layers is described below

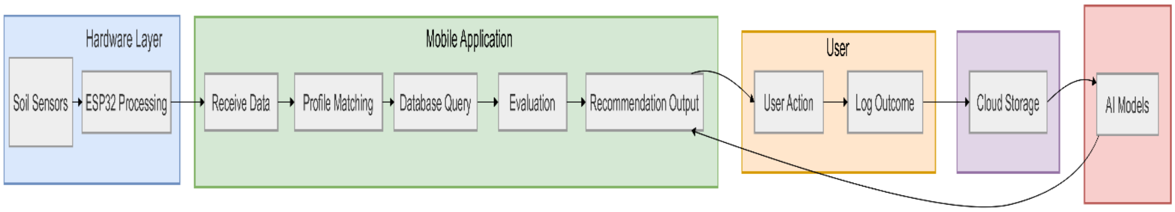


Figure 1-4 End-to-End System Architecture and Data Flow

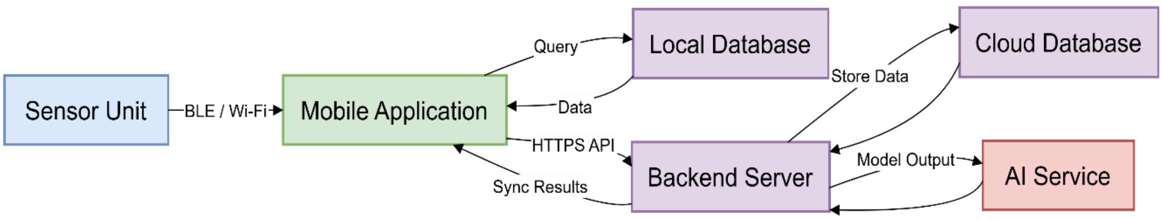


Figure 1-5 Component Interaction Diagram

1.2.2 Software Architecture and Technologies

a. Mobile Application Architecture

It follows a layered architecture pattern, separating the user interface layer, the business logic layer, and the data access layer into independent modules. This separation serves two purposes: it allows each layer to be tested, updated, and maintained independently, and it facilitates the integration of AI components.

Frontend (User Interface Layer)

The interface is built using React Native, React Native is selected over native Android (Java/Kotlin) or iOS (Swift) development for two reasons: it eliminates the need to maintain two separate codebases for the same interface, reducing development cost and ensuring function across platforms; and it provides a mature ecosystem for building data-display interfaces with real-time update behavior.

The interface is structured around five screens: profile setup, sensor activation and reading initiation, recommendation display, reading history, and application settings.

Business Logic Layer

Contains the recommendation engine — described in Section 1.1.2. — It receives sensor values and the active profile parameters from the interface layer, queries the database through the data access layer, applies the comparison logic, and returns a structured recommendation to the interface layer for display.

The business logic layer is extended with an AI module that receives outputs from the backend and incorporates them into the recommendation object. They both operate in parallel.

Data Access Layer

The data access layer manages all read and write operations to the local database. It provides the business logic layer with a clean query interface, abstracting the underlying database technology. This abstraction ensures that changes to the local database schema — necessary as the agronomic database expands — do not propagate to the business logic code.

b. Local Database

The database is implemented in SQLite, the embedded relational database engine standard for mobile applications. SQLite requires no server process, runs entirely within the application environment, and is supported natively on both Android and iOS.

The database schema is organized into three primary tables:

- **Crop Profiles:** Stores the agronomic reference data — optimal ranges for moisture, NPK, and pH indexed by crop type, growth stage, and climate zone. This table is read-only from the application's perspective; updates are pushed from the backend during synchronization.
- **User Profiles:** Stores user-defined crop profiles, including wilaya, selected crop type, and current growth stage.
- **Reading History:** Stores all sensor readings, their associated profile at time of reading, the recommendation generated, and the farmer-reported outcome for that recommendation cycle.

c. Backend Architecture

The backend is implemented as a RESTful API service built with Python (FastAPI framework). FastAPI is selected for its native support for asynchronous request handling, its automatic API documentation generation, and its compatibility with the Python-based machine learning libraries used in the AI layer. The API exposes three endpoint groups:

- **Sync**: Updates local CropProfiles when version-tagged requests indicate outdated data.
- **Ingestion:** Stores anonymized mobile readings and outcome records in the cloud.
- **AI Output:** Provides Phase 2 results (risk scores, anomalies, weights) for the mobile application.

The API manages profile updates, anonymized data collection, and AI-driven insights for the mobile display. It utilizes a PostgreSQL database and a batch-job training pipeline that updates models based on the rate of data accumulation.

d. AI Layer — Technical Specification

operates as a backend-side Python service using the scikit-learn library for the anomaly detection models, and recommendation refinement.

Precondition for AI activation: No AI model is trained or deployed until the outcome dataset contains a minimum of 5,000 complete reading-recommendation-outcome cycles across at least three wilayas and two crop categories. This threshold ensures that model training is not initiated on data volumes too small to produce reliable generalizations. The mobile application logs data from first deployment; the AI layer is activated when this threshold is reached, estimated at 18 to 24 months post-launch.

e. API and Data Exchange

Communication between the mobile application and the backend uses HTTPS with JSON-formatted payloads. All data exchanged is anonymized before transmission: user identifiers are replaced with hashed tokens, and geographic data is reduced to wilaya-level granularity. The synchronization process is initiated by the mobile application when a network connection is detected.

1.2.3 Infrastructure, Hosting and Security

a. Hosting Architecture

Algerian national cloud infrastructure (Algérie Télécom Cloud), where available and technically sufficient, satisfies data sovereignty requirements more directly.

b. Server and Hardware Configuration

A baseline configuration is defined as follows:

Application Server:

2 vCPU

4 GB RAM

Containerized deployment (Docker)

FastAPI framework

Database Server (PostgreSQL):

2–4 vCPU

8 GB RAM

SSD storage for efficient read/write operations

c. Security Strategy

Transmission level: All communication between the mobile application and the backend is encrypted using TLS 1.3. ensuring that the application only communicates with the authenticated backend server.

Backend level The REST API is protected by token-based authentication (JWT — JSON Web Tokens). Each mobile application installation generates a unique hashed device token at first launch, which is used to authenticate all subsequent API requests. No personal identifying information is transmitted to or stored on the backend.

d. Update Management

Application updates: (interface, business logic, AI inference module) are distributed

database updates: (new crop entries, revised parameter ranges) are delivered through the database sync endpoint without requiring an application update.

1.3 Development, Deployment and Technical Feasibility

1.3.1 Development Organization and Methodology

a. Development Methodology

AGRIGENT adopts a hybrid Agile-Lean Startup methodology as its development framework. For the system integrates hardware and software components with interdependent development timelines.

b. Development Phases

The project is organized into four sequential phases

Phase 1 — Prototype Development (current phase)
Scope: Hardware assembly and sensor validation; a web platform with core navigation, profile management, Bluetooth data reception, and rule-based recommendation engine.

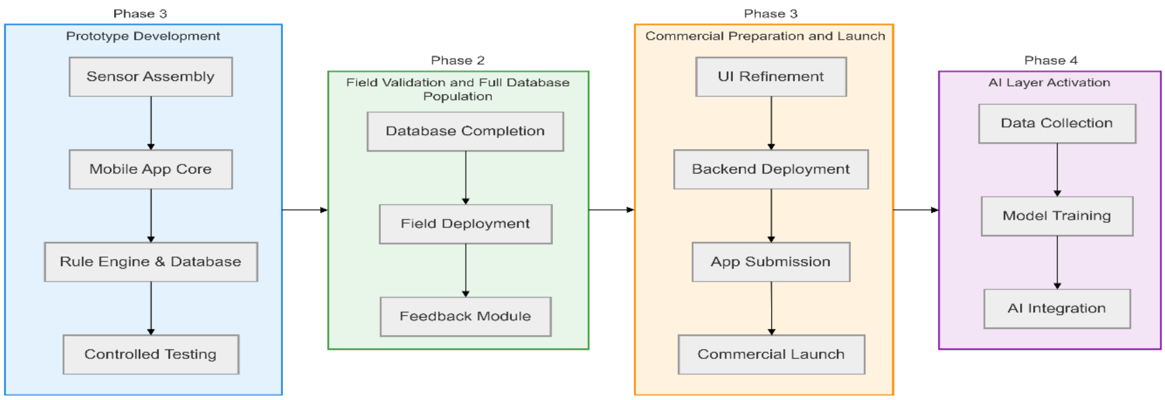
Phase 2 — Field Validation and Full Database Population
Entry condition: Prototype passes controlled test criteria.

Deployment of the prototype in real agricultural field conditions. Simultaneous completion of agronomic database population using INRAA and MADR reference data, complemented by FAO and CIHEAM sources as specified in Section 1.1.2.

Phase 3 — Commercial Preparation and Launch
User interface refinement based on field feedback; application deployed, backend production infrastructure deployment; pricing and distribution finalization.

Phase 4 — AI Layer Activation
Entry condition: Outcome dataset reaches the minimum threshold defined in Section 1.2.2, estimated at 18 to 24 months post-launch.
Development and training of the three AI models defined in Section 1.2.2 integration of AI outputs into the mobile application.

c. Development Planning



1.3.2 Validation, Deployment and Technical Limits

a. Functional and Technical Testing

Hardware testing: Each sensor is individually validated against reference measurements before assembly into the final unit.

Software testing: The recommendation engine is tested against a predefined set of input scenarios designed to cover all recommendation categories.

Integration testing: End-to-end testing validates the complete data flow described in Section 1.2.1: Integration tests are executed on the physical sensor unit connected to a test device, covering both the normal operation path and edge cases.

b. Field Validation Protocol

Technical reliability: The system must complete a sensor reading and generate a recommendation without error in at least 90% of activation attempts across 50 consecutive field test sessions.

Agronomic coherence: Recommendations generated by the system are reviewed by an agricultural engineer against the measured soil conditions. Agreement between the system output and independent expert assessment is the primary indicator of recommendation quality.

User comprehension: A structured observation session with five farmers — representative of the primary user profile — documents whether recommendation outputs are understood and acted upon correctly without assistance.

c. Deployment Modalities

Hardware distribution is managed through direct sales to farmers and agricultural investors, in target wilayas.

Application distribution follows standard mobile platform procedures — Google Play Store and Apple App Store. Application updates, agronomic database content updates described in Section 1.2.3.

initial commercial deployment targets three to five wilayas with the highest concentration of cereal and vegetable cultivation — consistent with the market segmentation established in Section 1.1.2 —.

d. Maintenance and Evolutionary Roadmap

Agronomic database maintenance: Parameter ranges in the CropProfiles database are reviewed on an annual basis against updated INRAA and MADR agronomic guidance. delivered to user devices as specified in Section 1.2.3.

Application maintenance: Bug fixes and security patches are distributed through standard platform update mechanisms. as described in Section 1.2.3.

Evolutionary development: The roadmap follows the phase structure defined in Section 1.3.1. Phase 4 AI layer activation is the primary evolutionary milestone. Intermediate additions — additional crop types, new climate zone entries— are delivered through database updates and minor application releases within the existing architecture, without requiring structural redesign.

e. Current Technical Constraints

Sensor accuracy under variable conditions: Electrochemical NPK sensors may show reduced accuracy under certain soil conditions. This is addressed through — an architectural decision documented in Section 1.1.2.

Connectivity limitations in rural areas: Network coverage may be intermittent or unavailable in agricultural zones.

AI activation: The 18 to 24 months estimate for reaching the AI activation threshold is based on conservative user adoption assumptions. If adoption is slower than projected, the threshold is reached later.

Rule-based decision limitations: The current recommendation engine operates on predefined agronomic rules and threshold ranges. While effective for structured decision support, this approach cannot capture complex interactions between soil parameters or long-term soil dynamics.

f. Technical Risks

Table 1-3 Technical Risks

Risk	Impact
NPK sensor calibration drift in field conditions	High
ESP32 Bluetooth range insufficient in dense crop environments	Medium
App store rejection during submission	Medium
Backend data loss during cloud provider outage	High
AI model underperformance at Phase 4 activation	Medium